Experiment 03 – Lab 4

Summarization and Text Transformation Report

Aim

To use ChatGPT to summarize text and transform its tone, comparing the original text with the generated output.

Theory

Summarization condenses a longer text into its main ideas while preserving its meaning. Text transformation rewrites content for a different tone or audience, such as changing an informal message into a formal one. The output should preserve the original facts and intended meaning.

Task 1: Summarization

Original text: Trees are important for the environment. They absorb carbon dioxide and release oxygen. Trees provide food and shelter for birds and other animals. Their roots help hold soil in place and reduce soil erosion. Trees also provide shade and help make cities cooler. Planting and protecting trees can improve air quality and support biodiversity.

Prompt used: Summarize the following paragraph in two sentences using simple English. Keep the main ideas and do not add new information.

ChatGPT output:

Trees help the environment by absorbing carbon dioxide, releasing oxygen, supporting wildlife, and reducing soil erosion. They also provide shade, cool cities, and help improve air quality and biodiversity.

Comparison: The output is shorter than the original while retaining its main ideas: air quality, wildlife, soil protection, shade, cooling, and biodiversity.

Task 2: Formal Tone Transformation

Original text (informal email/message): Hi, I need the project report today. Send it to me as soon as possible. Thanks.

Prompt used: Rewrite this message in a polite, professional tone. Keep the request and deadline unchanged.

ChatGPT output (formal):

Hello, could you please send me the project report by the end of today? I would appreciate receiving it as soon as possible. Thank you.

Comparison: The output is more polite and professional, while preserving the request for the report today.

Task 3: Informal Tone Transformation

Original text (formal): Dear Team, I would like to inform you that tomorrow's meeting has been postponed. The revised schedule will be shared shortly.

Prompt used: Rewrite this announcement in a friendly, informal tone without changing the facts.

ChatGPT output (informal):

Hi everyone! Tomorrow's meeting has been postponed. I'll share the updated schedule soon.

Comparison: The output sounds friendlier and more conversational while retaining the postponement and the promise to share a revised schedule.

Summary and Transformation Report

Task	Original	Output	Evaluation
Summarization	Long paragraph about environmental benefits of trees.	Two-sentence summary retaining the main points.	Main ideas preserved; wording shortened.
Formal transformation	Direct, informal request for a report today.	Polite, professional request with same deadline.	Tone improved; meaning preserved.
Informal transformation	Formal announcement about a postponed meeting.	Friendly, conversational announcement.	Tone changed; facts preserved.

Evaluation of Output

The summary keeps the main points while reducing length. The formal rewrite uses polite wording, and the informal rewrite uses a friendlier style. In all examples, the intended meaning is retained. These are illustrative examples; results can vary depending on the prompt and the AI response.

Conclusion

ChatGPT can summarize text and transform writing to suit different purposes and audiences. Clear prompts help control the length, tone, and style of the output. Comparing the original and rewritten text helps check that the important information and meaning have been preserved.

Result: Successfully prepared a summarization and text transformation report showing original text, prompts, generated outputs, and evaluation.